

Data Quality, Labeling, and Bias

Cleaning, annotation, imbalance, bias management, and training data integrity



Data quality has several dimensions

01 Completeness

Required records and fields are present at an acceptable level.

02 Validity

Values follow expected types, ranges, formats, and business rules.

03 Consistency

Definitions and values agree across files, systems, and time periods.

04 Freshness

Data arrives soon enough to represent current conditions and support the use case.



Cleaning decisions depend on meaning

Missing values



Remove, impute, or mark as unknown

The choice depends on why values are missing and how much data would be lost.

Duplicates



Identify the real unit of observation

Repeated events may be errors, retries, or valid repeated behavior.

Outliers



Investigate before removing

Extreme values can be errors, rare cases, or the most important examples.

Invalid values



Correct, map, quarantine, or reject

Document the rule so later data receives the same treatment.



Reliable labels require a controlled process

01

02

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04

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Define

Create clear label meanings, examples, and edge-case rules.

Assign

Use human reviewers, automated methods, or a combined workflow.

Review

Measure agreement and inspect uncertain or disputed examples.

Protect

Limit access and remove unnecessary sensitive information.

Version

Track guideline, dataset, and label changes over time.



Class imbalance can hide poor performance

The problem

When one class dominates, a model may achieve high overall accuracy while failing on the rare class that matters most.

Possible responses

Use suitable metrics, stratified splits, class weighting, resampling, augmentation, or better collection of minority examples.



Bias can enter at several points

01 Collection

The sample may exclude groups, locations, languages, devices, or situations.

02 Measurement

Proxies, sensors, surveys, or historical systems may record groups differently.

03 Labeling

Guidelines and reviewer judgment can create inconsistent or unfair labels.

04 Use and feedback

Model decisions can influence future data and reinforce earlier patterns.



AI training data needs integrity and safety checks

Traditional ML

Check schema, labels, leakage, duplicates, split quality, source permissions, and whether the data represents the intended population.

Generative AI

Validate prompt-response pairs, document provenance, content safety, licensing, privacy, harmful material, and multimodal distribution.



Trustworthy models begin with trustworthy data

Measure quality using rules connected to the use case.

Treat cleaning and labeling as documented decisions.

Evaluate minority classes and affected groups separately.

Track provenance, permissions, versions, and safety checks.

NEXT

BLA 3: Model and foundation-model development

